Math 261-001
Quiz 9 Solutions, November 6

1. **Final answers:**

$$\boxed{\text{(a) Line integral of } \vec{F}}, \qquad \boxed{\text{(b) Surface integral of } \operatorname{curl} \vec{G}}.$$

(a) Stokes' theorem replaces the flux of the curl through the hemisphere by the line integral of the original field along its boundary:

$$\iint_H (\operatorname{curl} \vec{F}) \cdot \vec{N}\, d\sigma = \oint_{\partial H} \vec{F} \cdot \vec{T}\, ds.$$

The boundary is the circle $x^2 + y^2 = 4$, $z = 0$. The outward orientation of the upper hemisphere induces counterclockwise traversal as viewed from above. Thus the replacement is a *line integral of* $\vec{F}$, not a line integral of its curl.

(b) Stokes' theorem replaces the line integral by the surface integral of the curl:

$$\oint_T \vec{G} \cdot \vec{T}\, ds = \iint_{S_T} (\operatorname{curl} \vec{G}) \cdot \vec{N}\, d\sigma.$$

One suitable surface S_T, with boundary T, is the flat triangular sheet in $x + y + z = 2$ with the three stated vertices. The traversal $(2,0,0) \to (0,2,0) \to (0,0,2) \to (2,0,0)$ induces an upward normal. Thus the replacement is a *surface integral of* $\operatorname{curl} \vec{G}$, not the flux of $\vec{G}$ itself.

Neither part requires differentiation or evaluation. The type of integral changes when passing to the other side of Stokes' theorem.

2. **Final answers:**

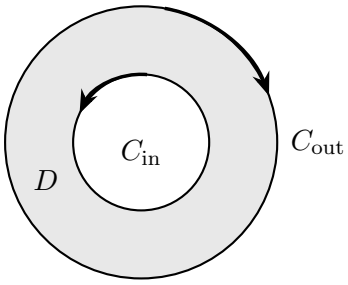
$$\boxed{\vec{c}_{\text{out}}(t) = (2\cos t, -2\sin t, 4)}, \qquad \boxed{\vec{c}_{\text{in}}(t) = (\cos t, \sin t, 1)}, \qquad 0 \le t \le 2\pi.$$

The paraboloid band has two boundary circles. At $z = 4$, the outer circle has radius 2; at $z = 1$, the inner circle has radius 1.

An outward normal, pointing away from the z-axis, is

$$\vec{n} = (2x, 2y, -1).$$

It has a negative z-component. Thus the projection is an annulus with *downward* orientation: the outer circle is **clockwise** and the inner circle is **counterclockwise**, viewed from above.



View from above; $N_z < 0$.

To check the normal directly, use

$$\vec{r}(r,\theta) = (r\cos\theta, r\sin\theta, r^2), \qquad 1 \le r \le 2.$$

Here $0 \le \theta \le 2\pi$. The outward cross product is

$$\vec{r}_\theta \times \vec{r}_r = (2r^2\cos\theta, 2r^2\sin\theta, -r).$$

The displayed boundary parametrizations have the required directions and the correct heights. The shaded annulus D is the xy-projection of the curved surface S.

3. **Final answer:** $\boxed{3}$.

Write the plane as $z = 2 - 2x - y$ and parametrize the sheet by

$$\vec{r}(x, y) = (x, y, 2 - 2x - y), \qquad \begin{cases} 0 \le x \le 1, \\ 0 \le y \le 2 - 2x. \end{cases}$$

Its tangent vectors give

$$\vec{r}_x = (1, 0, -2), \qquad \vec{r}_y = (0, 1, -1), \qquad \vec{n} = \vec{r}_x \times \vec{r}_y = (2, 1, 1).$$

The positive third component gives the required upward orientation. The unit normal is $\vec{N} = (2, 1, 1)/\sqrt{6}$ and $d\sigma = \sqrt{6}\, dx\, dy$, so $\vec{N}\, d\sigma = (2, 1, 1)\, dx\, dy$.

On the surface,

$$\vec{F}(\vec{r}(x, y)) = (y, x, y + 2 - 2x - y) = (y, x, 2 - 2x),$$

and hence

$$\vec{F}(\vec{r}) \cdot (\vec{r}_x \times \vec{r}_y) = 2y + x + 2 - 2x = 2 + 2y - x.$$

Therefore

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{N}\, d\sigma &= \int_0^1 \int_0^{2-2x} (2 + 2y - x)\, dy\, dx \\ &= \int_0^1 \left[(2 - x)y + y^2\right]_0^{2-2x} dx \\ &= \int_0^1 (8 - 14x + 6x^2)\, dx \\ &= \left[8x - 7x^2 + 2x^3\right]_0^1 = 3. \end{aligned}$$

4. **Final answer:** $\boxed{6}$.

For $\vec{F} = (z - x^{2026}, x - y^{2026}, y - z^{2026})$, the curl is

$$\operatorname{curl} \vec{F} = (1 - 0,\ 1 - 0,\ 1 - 0) = (1, 1, 1).$$

Choose the flat triangular sheet with vertices $P = (2, 0, 0)$, $Q = (0, 2, 0)$, and $R = (0, 0, 2)$. Since

$$(Q - P) \times (R - P) = (-2, 2, 0) \times (-2, 0, 2) = (4, 4, 4),$$

the stated traversal of T, $P \to Q \to R \to P$, matches an upward normal. Parametrize the triangle by

$$\vec{r}(x, y) = (x, y, 2 - x - y), \qquad \begin{cases} 0 \le x \le 2, \\ 0 \le y \le 2 - x. \end{cases}$$

Then $\vec{r}_x \times \vec{r}_y = (1, 1, 1)$. By Stokes' theorem,

$$\begin{aligned} \oint_T \vec{F} \cdot \vec{T}\, ds &= \iint_{S_T} (\operatorname{curl} \vec{F}) \cdot \vec{N}\, d\sigma \\ &= \int_0^2 \int_0^{2-x} (1, 1, 1) \cdot (1, 1, 1)\, dy\, dx \\ &= \int_0^2 \int_0^{2-x} 3\, dy\, dx = \int_0^2 (6 - 3x)\, dx = 6. \end{aligned}$$

The large powers disappear because only derivatives with respect to the other components' variables enter the curl.